

US Top 50 Music Analytics

Executive Summary Report

Prepared For:

Government Stakeholders & Senior Decision-Makers

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Table of Contents

1. Project Overview
2. Objective of the System
3. Problem Statement
4. Importance of Music and Ranking Analytics
5. Role of AI and Machine Learning in Music Intelligence
6. Dataset Overview
7. Data Processing and Feature Engineering
8. Song, Artist and Content Segmentation
9. Analytical Models and Performance Assessment
10. Decision-Support and Recommendation Overview
11. Key Findings and Insights
12. Analytical Performance Highlights
13. Benefits for Music Platforms and Government Bodies

14. Potential Real-World Applications

15. Scalability and Future Scope

16. Final Conclusion

1. Project Overview

The **US Top 50 Music Analytics** project is a data-analytics solution for understanding how songs, artists and content attributes behave within a daily US Top 50 music chart. It converts repeated chart snapshots into longitudinal measures of rank, movement, popularity, chart longevity and artist dominance.

The project covers the full descriptive analytics lifecycle: data inspection, duplicate handling, date and rank validation, feature engineering, song-level KPI construction, artist-level aggregation, correlation analysis and dashboard-oriented visualisation. The resulting Streamlit design supports playlist timeline exploration, ranking trends, artist dominance, popularity-versus-rank analysis and explicit-versus-non-explicit performance.

The project is an analytical and dashboarding solution. It does not claim supervised ML, clustering, SHAP or predictive model deployment where those methods were not present in the submitted notebook.

2. Objective of the System

- Explore how songs enter, remain on and move through the US Top 50 playlist over time.
- Measure song-level chart longevity, average rank, best rank and rank volatility.
- Identify artists with strong chart presence using unique-song breadth, total chart days and an Artist DominanceIndex.
- Examine the relationship between platform popularity scores and chart rank.
- Compare explicit and non-explicit content on popularity and rank outcomes.
- Assess album type, song duration and album size as contextual attributes associated with chart performance.
- Provide an interactive Streamlit dashboard with date, artist, song, rank-range and album-type controls.
- Create a reproducible analytical foundation for future forecasting and recommendation systems.

3. Problem Statement

Music charts generate repeated observations of the same songs, but simple daily rank tables do not reveal persistence, movement, artist concentration or the relationship between chart position and platform popularity. Without longitudinal analysis, decision-makers may miss which songs sustain attention, which artists dominate exposure, and how content attributes relate to performance.

The project addresses this gap by transforming daily Top 50 observations into song- and artist-level KPIs. It also separates entry/exit effects from consecutive-day rank movement and provides an interactive interface for exploring the resulting patterns.

4. Importance of Music and Ranking Analytics

- **Audience Insight:** longitudinal chart data helps platforms understand sustained versus short-lived audience attention.
- **Content Planning:** song longevity and category-level performance can support playlist and promotion decisions.
- **Artist Exposure:** dominance measures reveal how concentrated chart visibility is among leading artists.
- **Market Monitoring:** rank movement identifies rapid changes in audience interest.
- **Accountability and Measurement:** standard KPIs create repeatable measures for chart performance and platform engagement.

5. Role of AI and Machine Learning in Music Intelligence

The reference executive summary highlights AI/ML for automated pattern discovery, prediction, personalisation, continuous learning and evidence-based decisions. In the submitted music project, these capabilities are treated as **future extensions**. The completed work focuses on descriptive, longitudinal and correlational analytics.

The engineered dataset is nevertheless suitable for future forecasting tasks. Historical rank, popularity, movement, chart duration and artist-level features could support time-series prediction or recommendation models, provided the target definition, temporal validation and leakage controls are designed carefully.

6. Dataset Overview

The notebook starts with **22,994 rows and 10 columns** from the source file *Atlantic_United_States.csv*. The fields include date, position, song, artist, popularity, duration, album type, total tracks, explicit-content status and album-cover URL. After duplicate date-position records were removed, the analytical working dataset contains **22,944 rows**. The notebook later reports a final exported cleaned dataset of 27,750 rows and 19 columns; this difference is documented as a pipeline/reproducibility discrepancy rather than silently reconciled.

Dataset / Indicator	Observed Value
Original dataset	22,994 rows × 10 columns
After duplicate date-position removal	22,944 rows
Rank range	1–50; no invalid ranks reported
Unique artists	273
Unique songs in song KPI table	861
Main analytical entities	Songs, artists, daily chart positions
Core popularity metric	Popularity score
Final exported file reported by notebook	27,750 rows × 19 columns

7. Data Processing and Feature Engineering

The notebook performs duplicate detection, date parsing, rank validation and construction of longitudinal features. It identifies duplicate song-date observations, with 96 duplicates concentrated on 01-03-2025, and removes duplicate date-position records. It also documents a malformed date value and an unusual daily record count as data-quality anomalies.

Processing Area	Approach
Duplicate handling	Remove duplicate date-position records; working dataset becomes 22,944 rows
Date processing	Convert date to datetime and inspect malformed/anomalous dates
Rank features	previous_rank, rank_movement, movement_type, entry/exit rank
Temporal features	previous_date, day_gap, daily_rank_movement
Song KPIs	days_on_chart, average_rank, best_rank, rank_volatility
Popularity features	average popularity and popularity trend score
Artist KPIs	unique-song breadth, total chart days, dominance index
Content analysis	explicit status, album type, duration and album size

8. Song, Artist and Content Segmentation

The project uses **analytical segmentation** rather than machine-learning clustering. Songs are grouped by longevity, rank performance, popularity and volatility; artists are compared by unique-song breadth, total chart days and dominance; content is compared by explicit status and album type.

Segment / Metric	Key Observation
Longest chart presence	Something in the Orange — 536 days
Second longest	Last Night — 492 days
Third longest	I Remember Everything — 446 days

Top artist by unique-song breadth	Taylor Swift — 96 unique songs
Top artist by total chart days	Zach Bryan — 546 days
Top Artist Dominance Index	Taylor Swift — 7.26%
Highest popularity example	un x100to — average 99.20, 79 days

9. Analytical Models and Performance Assessment

No predictive machine-learning model was trained in the submitted music notebook. Accordingly, classification accuracy, precision, recall, ROC-AUC, clustering scores and SHAP explanations are not claimed. The project's performance assessment is based on descriptive and statistical measures that directly correspond to chart behaviour.

The principal measures are average rank, best rank, rank volatility, days on chart, average popularity, popularity trend score, artist dominance and correlations between chart and content variables. These measures provide transparent evidence for comparing songs, artists and content groups.

10. Decision-Support and Recommendation Overview

The Streamlit dashboard functions as a **music intelligence and decision-support interface**. Its required modules include a playlist timeline explorer, song ranking trends, artist dominance leaderboard, popularity-versus-rank scatter analysis and explicit-versus-non-explicit performance panels. User controls include date range selection, artist and song filters, a rank-range slider and album-type selection.

A future recommendation engine could combine chart momentum, sustained popularity, artist exposure and user preferences to rank songs or playlists. Such a system would require user-level listening data and careful temporal validation; chart data alone should not be interpreted as an individual user's preference.

11. Key Findings and Insights

- **Chart Longevity:** Something in the Orange has the longest observed chart presence at 536 days, followed by Last Night at 492 days.
- **Artist Breadth:** Taylor Swift has the highest unique-song breadth with 96 songs represented in the song-level KPI table.
- **Artist Dominance:** Taylor Swift leads the Artist Dominance Index with 2,015 appearances, representing 7.26% of chart appearances.
- **Popularity and Rank:** the popularity-versus-rank correlation is weakly negative at -0.094, indicating that popularity score alone does not strongly determine chart position.
- **Rank-Group Popularity:** average popularity is 89.61 for ranks 1–10, 87.59 for ranks 11–20 and 87.03 for ranks 21–50.
- **Explicit Content:** explicit tracks represent 47.94% of observations and have average popularity of 87.30 versus 87.99 for non-explicit tracks.
- **Album Type:** singles show the highest average popularity at 90.13 and average rank of 23.99, while compilations perform lower on both measures.
- **Duration Effects:** song duration has only weak correlations with popularity (-0.043) and rank (-0.045).
- **Popularity Volatility:** popularity volatility and rank volatility show a modest positive correlation of 0.16.

12. Analytical Performance Highlights

Indicator	Result
Average Days on Chart	28.4

Overall Average Rank	30.05
Average Rank Volatility Index	7.71
Average Popularity Trend Score	75.66
Top Artist Dominance Index	Taylor Swift — 7.26%
Explicit Content Share	47.94%
Non-explicit Share	52.06%
Popularity vs Rank correlation	-0.094
Popularity volatility vs Rank volatility	0.16
Album type with highest average popularity	Single — 90.13

13. Benefits for Music Platforms and Government Bodies

For Music Platforms and Industry Stakeholders

- Improved playlist and catalogue monitoring through longitudinal rank analysis.
- Identification of durable versus short-lived chart performers.
- Artist exposure measurement through dominance and breadth metrics.
- Evidence-based promotional planning using popularity, rank movement and longevity.

For Government Bodies and Policymakers

- Evidence for understanding digital cultural consumption patterns at an aggregated level.
- Support for cultural-industry and creative-economy monitoring.
- Transparent indicators for studying concentration of platform visibility among artists.
- Potential inputs for digital-media policy and public cultural analytics, subject to licensing and privacy requirements.

14. Potential Real-World Applications

Application Area	Potential Use
Streaming / Playlist Platforms	Monitor chart momentum, song longevity and artist dominance.
Record Labels	Compare artist exposure, sustained performance and promotional opportunities.
Music Discovery Systems	Use longitudinal popularity and rank signals as inputs to recommendation models.
Application Area	Potential Use
Cultural-Economy Analytics	Track aggregated patterns in digital music consumption.
Media and Entertainment Research	Study artist concentration, content attributes and market movement.
Dashboard-Based Monitoring	Provide interactive exploration of chart trends for analysts and decision-makers.

15. Scalability and Future Scope

- Extend the dataset to longer historical periods and additional charts or geographic markets.
- Automate scheduled ingestion and quality checks for new daily chart snapshots.
- Build time-series forecasting models for rank and popularity using leakage-safe temporal validation.
- Develop recommendation models using user listening behaviour in addition to chart-level signals.

- Add NLP analysis of song metadata, lyrics summaries where legally permitted, genres and editorial descriptions.
- Introduce artist-network analysis for collaborations, co-appearances and cross-genre reach.
- Add explainable AI tools for future predictive models and monitor model drift as audience behaviour changes.

16. Final Conclusion

The US Top 50 Music Analytics project demonstrates how repeated chart snapshots can be transformed into a longitudinal intelligence system for songs, artists and content attributes. By engineering rank movement, chart longevity, volatility, popularity and artist dominance measures, the project moves beyond a simple daily Top 50 table and provides a structured view of market behaviour.

The findings show that chart success is multidimensional: long chart presence, high popularity, strong rank position and artist dominance do not reduce to a single metric. The weak popularity-versus-rank correlation further supports the need for multi-factor analysis.

For government and senior stakeholders, the project offers a transparent framework for aggregated digital-culture monitoring and creative-economy analytics. Future predictive and recommendation capabilities can be added once richer longitudinal and user-level data are available, with appropriate validation, licensing, privacy and fairness safeguards.

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Technology focus: Python · Pandas · NumPy · Streamlit · Data Visualisation